NEEDLE AGENT

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Score: 79/150 (52.7%) on the public test runner

*A self-restarting, watchdog-guarded, multi-tier coding agent that

overnight ground a knitting compiler out of nothing.*

the rules of the game

- Read a hidden spec at 20:00 Thursday
- Build the program it describes
- Submit by 12:00 Friday
- Use only **free** or **local** models — no paid Claude/GPT/Copilot
- Judges grade: 40% hidden tests + 20% public + 20% process + 10% self-tests + 10% talk
- The agent is the project. The program is just the battlefield.

what we built (before reveal)

A TypeScript/Bun coding agent inspired by the leaked Claude Code source.

- **9 tools**: `read`, `edit`, `write`, `bash`, `glob`, `grep`, `run_tests`, `load_skill`, `spawn_subagent`, `submit_done`
- **`edit` invariants from Claude Code**: read-before-edit, exact-match, mtime guard
- **query/tool-call loop** with cycle detection, auto-compaction, XML-tool-call recovery
- **Multi-tier model router** with timeout, retry, 429 cooldown, fallback chain
- **Subagent system** (planner / implementer / tester / failure_analyst / self_test_writer)
- **10 pre-written skills** + 2 knitting-domain skills (`knitting-error-format`, `knitting-stitch-math`)
- **Pretty terminal UI** with ANSI boxes (Claude Code vibe)
- **Watchdog** that restarts agent on death and restores best `knit.py` on regression

the reveal (20:00)

`secret_spec/SECRET_SPEC.md` — **Knitting Compiler**, 927 lines

```
python3 solution/knit.py compile <file.knit>
```

Parses a knitting DSL, validates, expands repeats, simulates stitch counts row by row, emits one deterministic JSON. 150 public tests across 8 levels of difficulty.

the journey

TIME	SCORE	WHAT
20:30	3/150	first scaffold via Featherless Qwen3-Coder-480B
22:16	51/150	basic parsing + simulation
00:00	76/150	cloud peak — then regressed by an over-eager edit
03:00	71/150	local Ollama overnight grind
09:54	77/150	Cerebras Cloud Qwen3-235B back in primary
10:20	**79/150**	knitting-domain skills + planner workflow

the model carousel

TRIED	VERDICT
DeepSeek V4-Pro (Featherless)	`tools=false` — useless
DeepSeek V3.2 (Featherless)	empty-args bug
DeepSeek V3-0324 (Featherless)	markdown instead of native tools
Qwen3-Coder-480B (Featherless)	slow + drops
Qwen3-Coder-480B (NVIDIA NIM)	rate-limited fast
Qwen3-235B (**Cerebras free**)	~2000 tok/s — primary
Qwen3-Coder-30B UD-Q4_K_XL (local Ollama)	~80 tok/s — fallback
Qwen3-Coder-30B IQ4_NL (local)	broken — invented tool names like `_write`, args like `"Eight"`

the overnight grind (in numbers)

METRIC	COUNT
agent decisions logged	**906**
commits since checkpoint	**1187**
watchdog auto-restarts	23
Cerebras-cooldown events	808
thinking-without-action nudges	621
context auto-compactions	199
provider auto-fallbacks	66
cycle prunes	11
all-providers-down events	4
run_tests invocations	42

what worked

- **Per-iteration auto-commit** — `git log` is a play-by-play of the agent's thinking
- **Claude Code's edit invariants** — read-before-edit + exact-match killed an entire class of silent overwrites
- **Cycle detector** — 3+ identical tool calls in a 5-window → context prune
- **Auto-compaction** — old tool results squashed to head+tail when context > 60 KB
- **3-tier model failover** + 429 cooldown — kept the loop alive when Cerebras hit TPM
- **Watchdog with regression restore** — pulled the best `knit.py` back from commit history when the model overwrote it with garbage

what failed

- **Free-tier rate limits** stalled us 30 min at a time when all 3 cloud providers were 429
- **IQ4_NL quantization** of Qwen3-Coder produced hallucinated tool calls — switched to **UD-Q4_K_XL**
- **Ollama registry** had EOF-on-manifest issues for `qwen3-coder:30b` — bypassed via `hf.co/unsloth/...:UD-Q4_K_XL`
- `submit_done` **premature calls** at sub-150 scores → tightened prompt to require ≥ 140
- **Level_05 (single errors) only 2/30** — knitting error JSON format was the biggest unsolved bucket

what we'd improve given more time

- A `verify_score` **tool** that wraps `run_tests` and lets the loop refuse `submit_done` when score < 140
- **High-water-mark snapshots inside the agent** (today we did it via an external watchdog)
- **Bigger local quant** (Q5_K_M / Q6_K) on hardware with more RAM
- **Spec-derived self-tests** generated proactively, not just at the end
- **Per-level subagents** that own one category and don't context-pollute the rest

the agent is the project

```
agent-readiness-1945  ← Thursday 19:45 checkpoint (set before the spec was even open)
              └─ 1187 commits ─→ HEAD  (every iteration a commit, every commit a decision)
```

Public score is the battlefield score.

The agent is what we built.

https://github.com/HackatonWinnners/Agent007

thank you 🧶